Solar Power for IoT Systems

Why Solar for IoT?

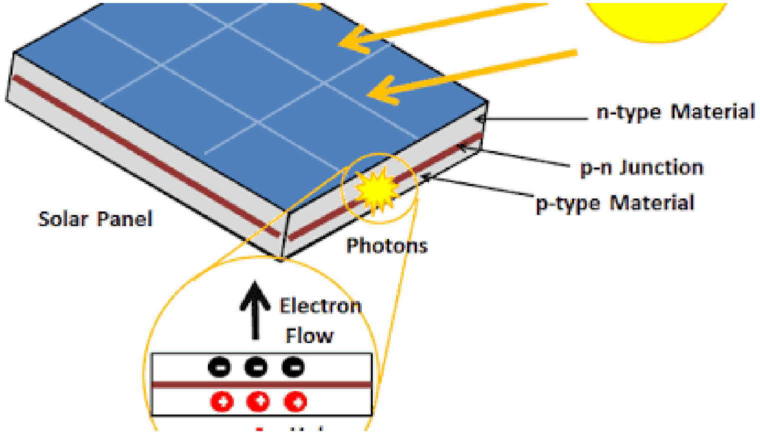
The dream of IoT: **deploy anywhere, run forever.**

```
Solar panel
    │  (free energy from sun)
    ▼
Charge controller
    │
    ├──► Battery (store for night/clouds)
    │
    └──► IoT device (sensor, MCU, radio)
```

> Goal: Generate enough during the day to run 24/7.

How a Solar Cell Works — The Photovoltaic Effect

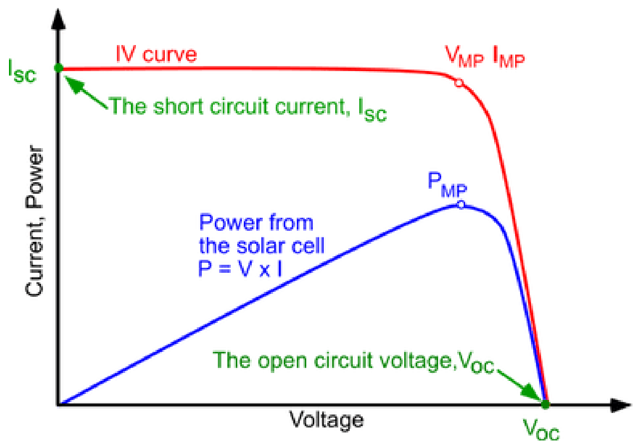
A solar cell converts **photons (light) → electrons (electricity)**.



1. Photon hits silicon → knocks electron loose
2. P-N junction creates electric field → pushes electrons one way
3. Electrons flow through external circuit → **DC current**
4. We can think of the solar cell as a **current source** with a voltage limit.

> One silicon solar cell produces ~**0.5–0.6V** regardless of size.
> Size determines **current** (more area = more photons caught).

Solar Cell — Voltage vs. Current



Parameter	Meaning
Voc	Open-circuit voltage — max voltage when no load
Isc	Short-circuit current — max current when shorted
Vmp	Voltage at maximum power point
Imp	Current at maximum power point
Pmax	Peak power = Vmp × Imp

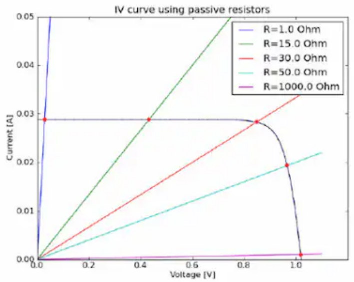
Key Solar Panel Specs to Know

```
Example panel label:
  Pmax  = 10W
  Vmp   = 17.6V      ← operating voltage at max power
  Imp   = 0.57A      ← operating current at max power
  Voc   = 21.6V      ← open-circuit voltage
  Isc   = 0.62A      ← short-circuit current
  Cell type: Monocrystalline
```

I-V Curve and Maximum Power Point (MPP)

- The Solar cell functions as the current source, but when it passes the certain voltage (Voc), the current drops to zero.
- The maximum power point (MPP) is the point on the I-V curve where the product of current and voltage is maximized.
- This is the optimal operating point for the solar cell to deliver

Adding a register to the solar cell:



- When a load (the register) is connected to the solar cell, it creates a voltage drop across the load, which affects the current output of the solar cell.
- When the load is too high (open circuit), the voltage is at Voc and current is zero.
- When the load is too low (short circuit), the current is at Isc and voltage is zero.

Adding a battery 3.8V to the solar cell:

- Battery functions as the voltage source, so the solar cell will try to charge the battery until it reaches the battery voltage.
- If the battery voltage is below Voc, the solar cell will provide current to charge the battery.
- If the battery voltage exceeds Voc, the solar cell cannot provide current and the battery will not charge.

Solar Panel Construction — Cells in Series & Parallel

Single cell: ~0.5V, small current

Combine cells to get useful voltage and current.

```
Series connection (voltage adds):
  [cell] → [cell] → [cell] → [cell]
  0.5V      0.5V     0.5V     0.5V  = 2.0V total                  .

Parallel connection (current adds):
  [cell]
  [cell]  ─── same voltage, 4× current
  [cell]
  [cell]
```

Typical IoT solar panels:

Panel	Voltage	Current	Power
Tiny (hobby)	5–6V	60mA	0.3W
Small (sensor)	6V	333mA	2W
Medium (gateway)	12V	830mA	10W
Large (off-grid)	18V	5.5A	100W

Product 1

- Solar Panel: Monocrystalline 0.5W 100mA
- Current: 100mA
- Connection: 2mm JST connecter

- The voltage is about 5V (0.5W/100mA = 5V), which is typical for a single solar cell or a small panel.
- However, it doesn't specify the Vmp or Voc.
- ESP32 needs 300 - 500 mA to run, so this panel is not sufficient to power an ESP32 directly.

Product 2

- Solar Panel: 10W Solar
- Voltage: 5V

- We can calculate the current: I = P/V = 10W/5V = 2A, which is much higher than the first panel.
- This panel can easily power an ESP32 and charge a battery at the same time.

Product 3

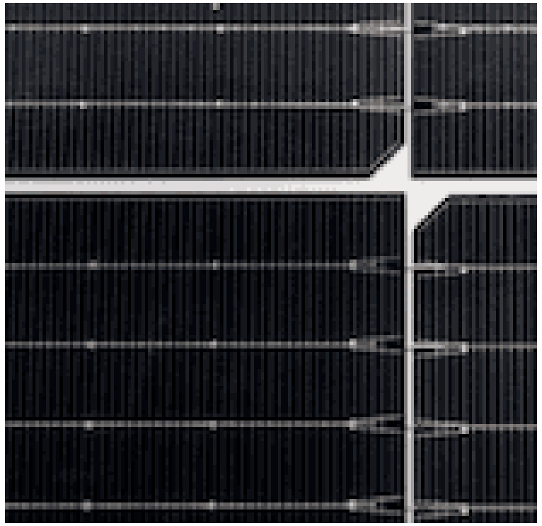
- Solar Panel: 10W Solar
- Voltage: 6V (12 high efficiency monocrystalline SunPower cells)
- Open Circuit Voltage: 8.51V
- Peak Voltage: 7.28V
- Peak Current: 330mA
- Peak Power: 2.37W

> monocrystalline -> 22+% efficiency

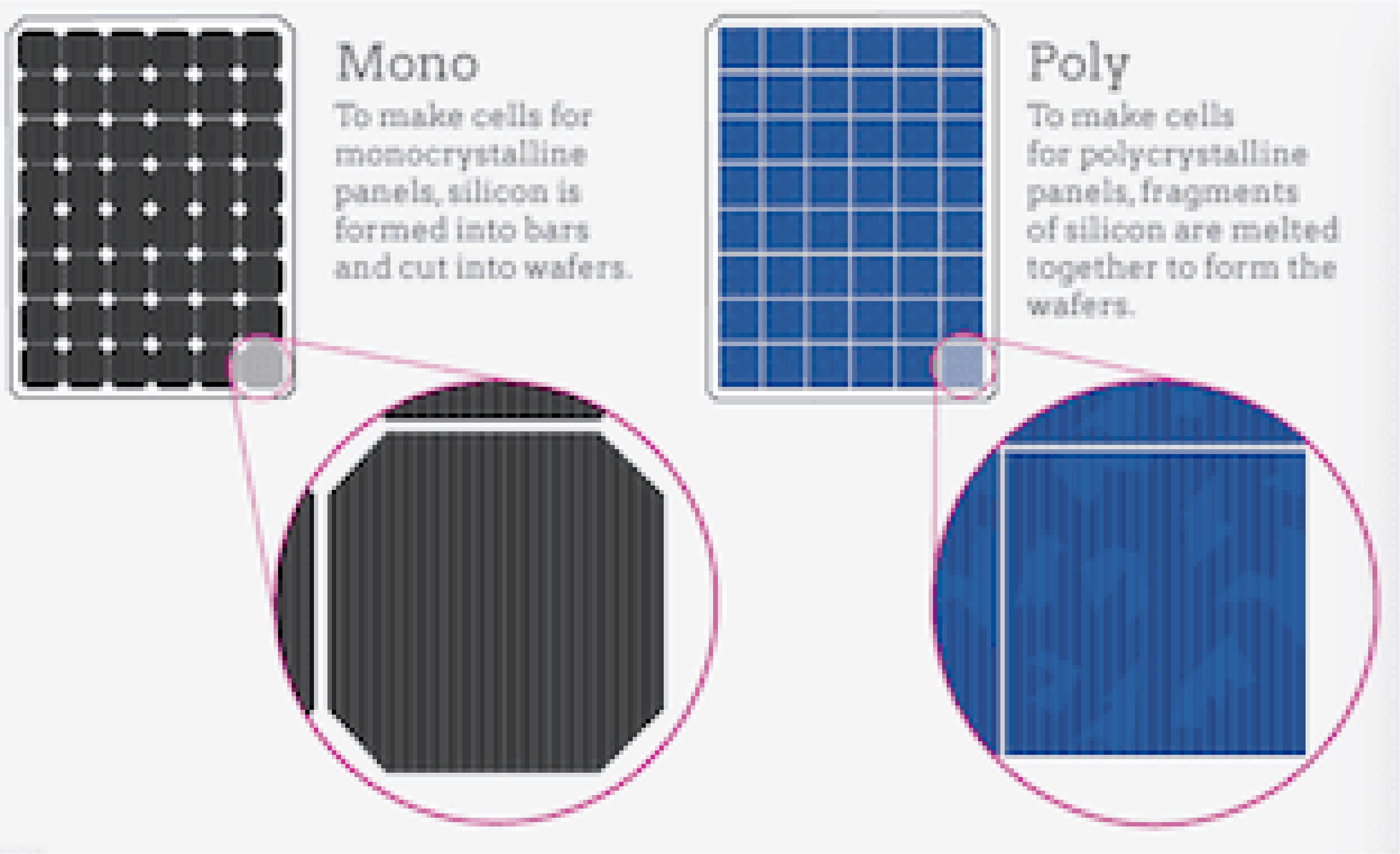
Cell Technologies:

Type	Efficiency	Cost	Best Use
Monocrystalline	20–22%	High	Space-limited IoT
Polycrystalline	15–17%	Medium	General outdoor
Amorphous (thin film)	7–10%	Low	Flexible/indoor

In most cases, we use **Monocrystalline** solar cells for IoT applications because they have the highest efficiency and are more compact, which is ideal for space-constrained IoT devices.



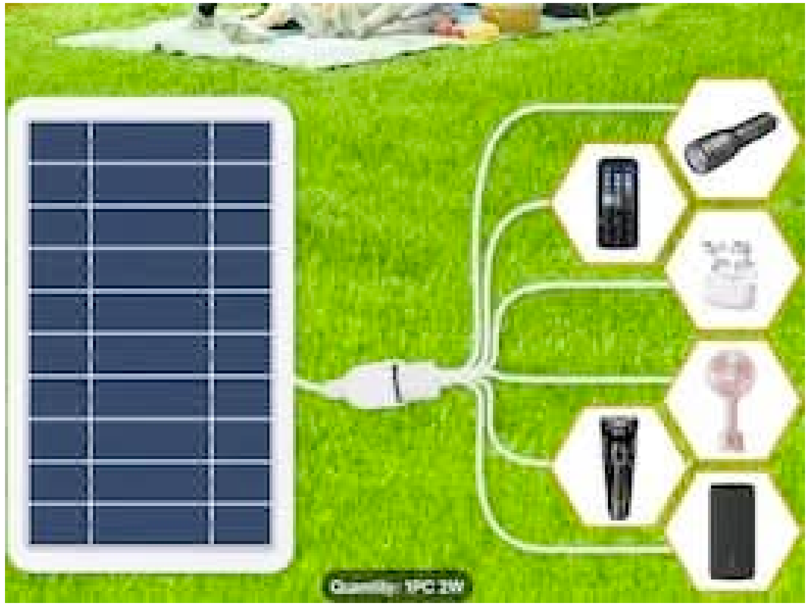
Before, Polycrstalline solar cells were more common because they were cheaper, but now the price gap has narrowed.


Mono
To make cells for monocrystalline panels, silicon is formed into bars and cut into wafers.
Poly
To make cells for polycrystalline panels, fragments of silicon are melted together to form the wafers.

For small gadgets, such as a calculator, we can use Amorphous solar cells because they are flexible and can be integrated into the device itself.

Power bank or Solar Panel Only?

Some solar panels are designed to charge batteries directly, but this is not ideal for IoT devices that need a stable voltage and current.



Use a solar power bank with a built-in charge controller to ensure safe charging and stable 5V output for your IoT device.

> Or design a custom solar charging circuit with a charge controller IC for higher efficiency and application-specific optimization.

Solar Charge Controllers

1. **PWM (Pulse Width Modulation) Controller**
    - Simple, cheap
    - Panel voltage must match battery voltage (e.g., 12V panel → 12V battery)
    - Wastes potential energy above battery voltage


```
Solar 17.6V → [PWM] → Battery 12V
               (chops voltage down, loses ~32% energy)
```

2. MPPT (Maximum Power Point Tracking) Controller

- More complex, more expensive
- Converts panel voltage to optimal battery charging voltage
- Extracts 10–30% **more energy** than PWM

> **For IoT:** small MPPT ICs (CN3791, BQ24650) are available in chips.

Mode1	Max. Solar Input Power		Max. DC Input Voltage	
	12V System	24V System	12V System	24V System
20A	240W	480W	50V	100V
30A	360W	720W	100V	100V
40A	480W	960W	100V	100V
60A	720W	1440W	100V	100V

```
Solar 17.6V × 0.57A = 10W → [MPPT] → Battery 12V × 0.8A = 9.6W
                                              (96% efficient conversion)
```

MPPT Idea

Problem: Solar panel's I-V curve shifts with temperature and light. The optimal operating point (Vmp) changes constantly.

```
Bright sun:               Cloudy day:
Vmp = 17.6V               Vmp = 15.2V
```

If the system keeps operating the panel at 17.6V, then:

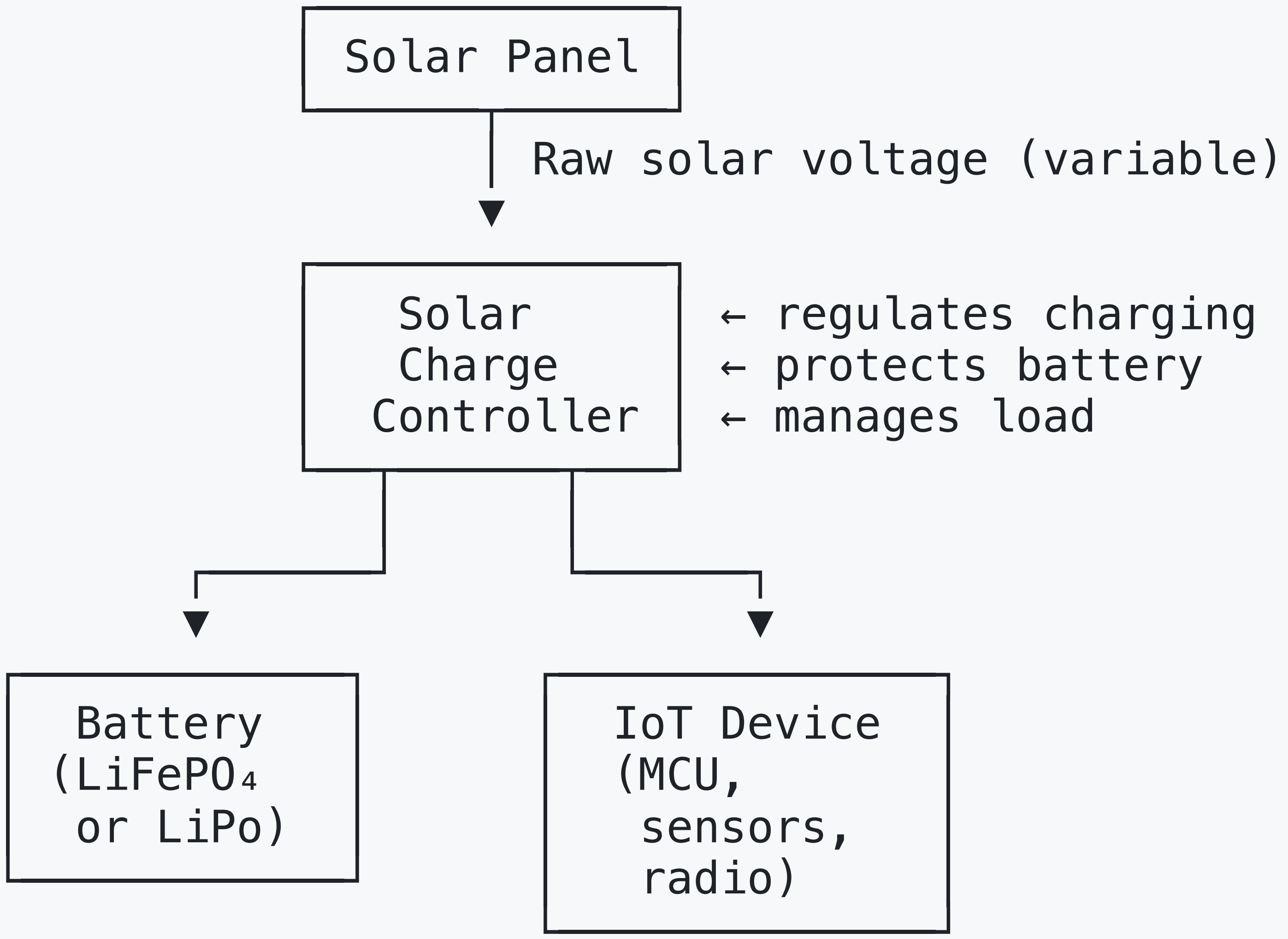
- On a bright day → it is near Vmp → maximum power.
- On a cloudy day → Vmp has shifted to 15.2V.
- The panel is no longer at its optimal point.
- Result → less power is extracted.

MPPT algorithm (Perturb & Observe):

```
1. Measure panel power P = V × I
2. Slightly increase panel voltage → measure new P
3. If P increased → keep going in same direction
   If P decreased → reverse direction
4. Repeat every ~100ms
→ Always tracks the peak power point
```

> MPPT controllers are essential for maximizing energy harvest.

Making a solar-powered battery charger for IoT devices



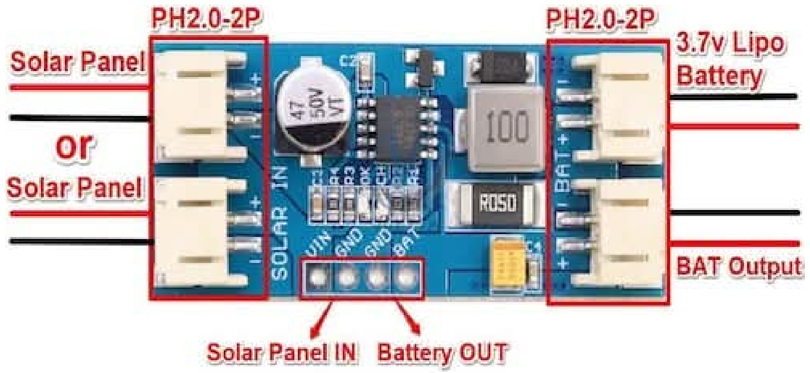
The **charge controller** is the brain of the system.

Solar Charge Controller ICs for IoT

Chip	Type	Battery	Solar Input	Features
CN3791	MPPT (fixed)	LiPo/Li-ion	Up to 28V	Simple, cheap ~$0.60
BQ24650	MPPT	Li-ion/LFP	Up to 28V	TI, high efficiency
SPV1040	MPPT	Any	Up to 5.5V	STMicro, tiny panels
AEM10941	MPPT	LiPo	Up to 4.5V	Ultra-low power IoT
CN3722	MPPT	Lead-acid	Up to 28V	For larger systems

CN3791 — Popular choice for small IoT solar systems:

```
Solar panel ──► CN3791 ──► LiPo battery
                     │
                   MPPT automatically maximizes power extraction
```



- Input voltage: 4.5V to 28V
- Max charge current: 1A
- Fixed MPPT tracking for optimal solar power harvesting
- Cost: ~$0.60 in bulk

Warning: CN3791 is a Battery Charger

CN3791 is designed to:

- Charge a 1S Li-ion / LiPo battery
- Perform simple fixed-ratio MPPT
- Control charging current

It is **NOT** a power path management IC.

Basic Architecture

```
Solar Panel → CN3791 → LiPo Battery → 3.3V Regulator → ESP32
```

- CN3791 only manages battery charging.
- ESP32 draws power from the battery.

Can ESP32 Run While Charging?

- Technically: Yes
- But there are limitations.

The Problem

1. High Current Spikes

ESP32 WiFi transmission: 400–500 mA peak current

If solar input is weak:

- Charging current drops -> Battery voltage dips -> ESP32 may reset

2. Low Sunlight Condition

If: ESP32 consumption > Solar charging current
Then:

- Battery continues discharging -> System eventually shuts down

When It Works Well

CN3791 is suitable when:

- ESP32 uses deep sleep
- Average current is low
- Solar panel has sufficient power margin

Ideal for low-power IoT nodes.

Better Solution for Stable Operation

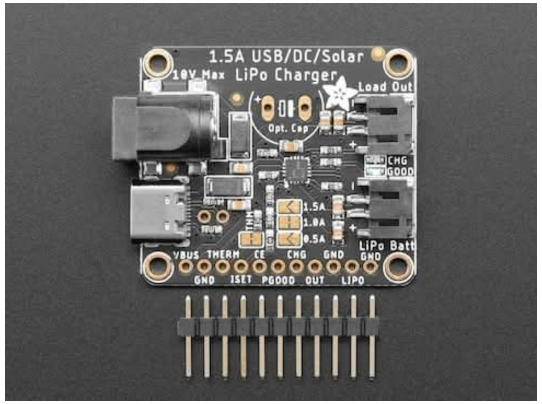
Use a Power Path Management IC, such as:
- BQ24074
- MCP73871

These allow:

```
Solar → System Power + Battery Charging (managed)
```

System power remains stable while charging.

BQ24074 — Power Path Management IC



- Up to 1.5A charge rate
- Up to 10V input voltage, 28V protected
- Load output is regulated to never be over 4.4V so it's safe for 3.3V regulators or 5V boost converters

However...

- There is no 'Done' LED - when the CHG LED turns off, that's how you know charging is complete
- Not 'true' MPPT, but near identical perforamnce.

1. Pickup any 3.7V/4.2V Li-ion battery.
2. Connect solar cell to the BQ24074 input.
3. Connect circuit load (ESP32) to the BQ24074 output.